4-Bit Carrylookahead Adder

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***Abstract*— This project explains the design and working of a 4-bit Carry Lookahead Adder (CLA). It adds two 4-bit binary numbers using full adders with carry lookahead logic. The circuit generates all carry signals in parallel using generate and propagate functions. Transient analysis was carried out to verify correct timing and performance.**

***Keywords*— Carry Lookahead Adder, Generate, Propagate, Parallel Carry, Transient Analysis**

I. Introduction

A 4-bit Carry Lookahead Adder is a digital circuit that performs the addition of two 4-bit numbers along with a carry input. Unlike a ripple-carry adder, where the carry moves step by step, the CLA computes carry signals in parallel. This is achieved using generate ($gi = ai \cdot bi$) and propagate ($pi = ai \oplus bi$) signals. The result is faster addition with reduced delay.

II. Objectives

The aim of this project is to design a 4-bit Carry Lookahead Adder that can add two 4-bit numbers with a carry input. The focus is on showing how generate and propagate signals allow carries to be calculated directly using lookahead logic. The objective is to build the truth table, run transient analysis in simulation, and confirm that the CLA works as expected.

III. Implementation

The 4-bit Carry Lookahead Adder is implemented using four full adders along with additional carry lookahead logic. Each full adder produces the sum bits, while the lookahead unit computes all the carry outputs in parallel based on the input bits. Instead of allowing the carry to move one stage at a time, the lookahead block directly determines whether a carry is generated or propagated at each level. This parallel operation makes the circuit faster and more efficient than ripple-based designs, while still maintaining correct binary addition for all possible input combinations.in eSim.

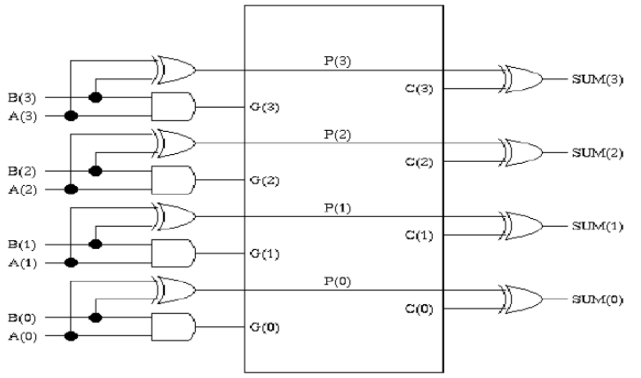

Fig. 2. Logic Diagram

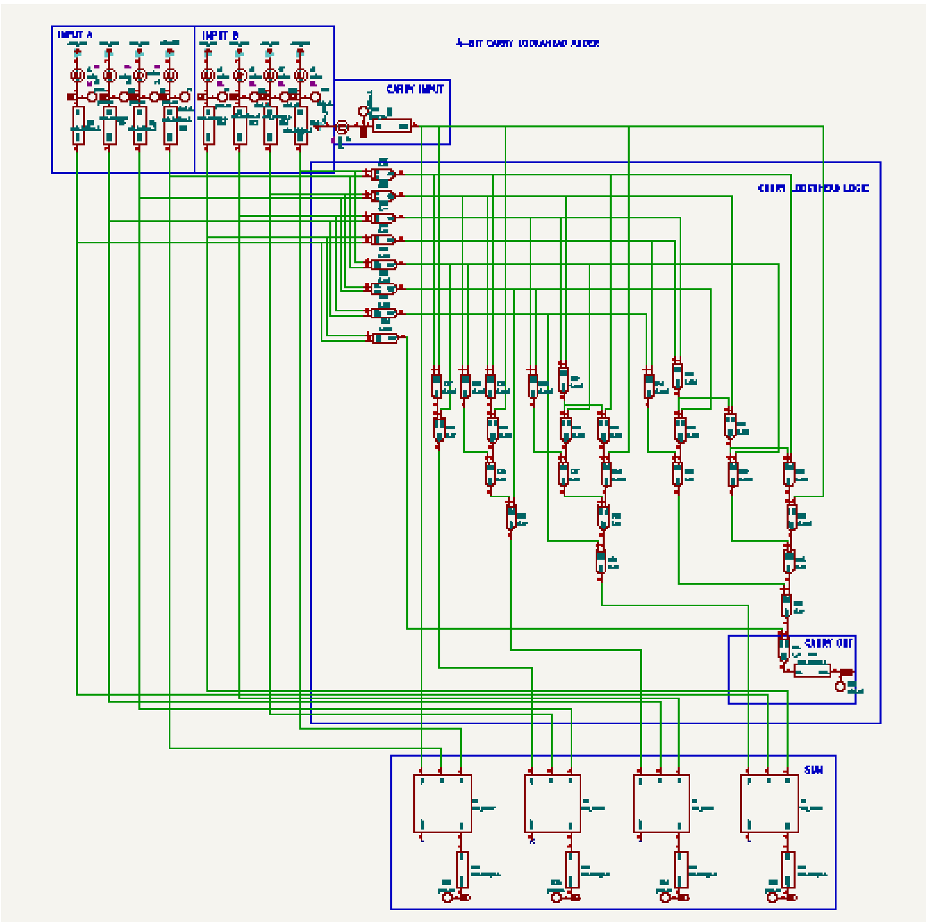

Fig. 1. eSim Circuit Schematic

IV. RESULTS

The 4-bit Carry Lookahead Adder was implemented and tested. The truth table was prepared with inputs ranging from 1111 to 0000, and the outputs of sum (S0–S3) and carry (C4) were verified. Transient analysis was performed to check the timing behavior. The results confirmed correct binary addition and showed that the carry signals were computed instantly through lookahead logic.

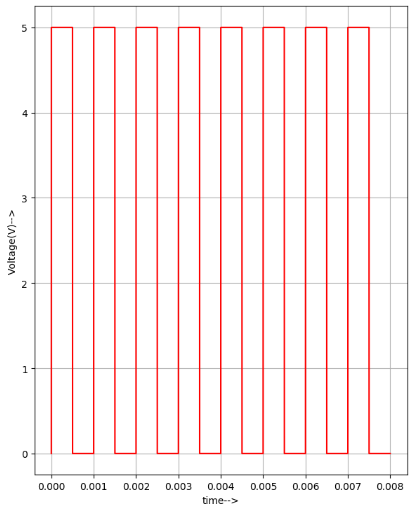

Fig. 3. Input A0

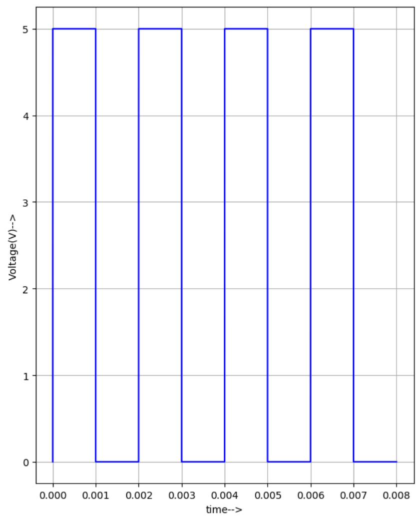

Fig. 4. Input A1

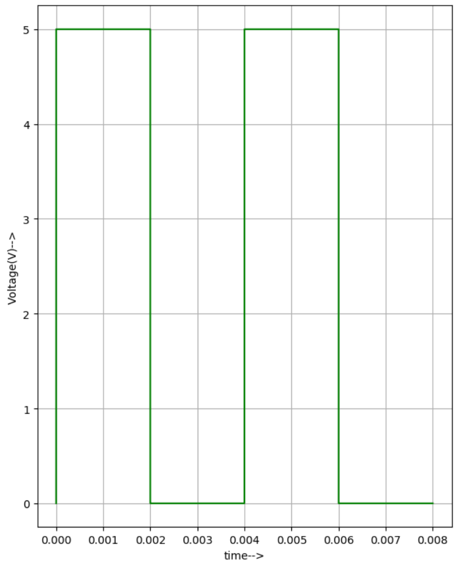

Fig. 5. Input A2

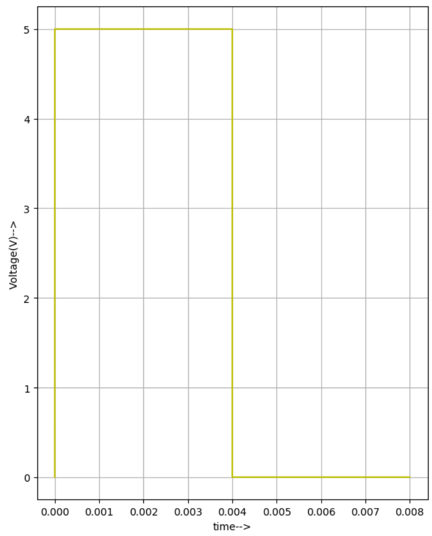

Fig. 6. Input A4

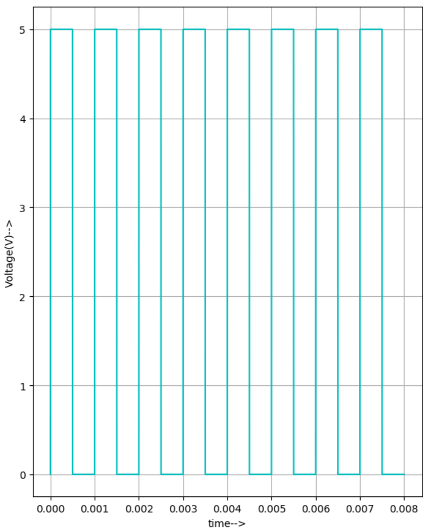

Fig. 7. Input B0

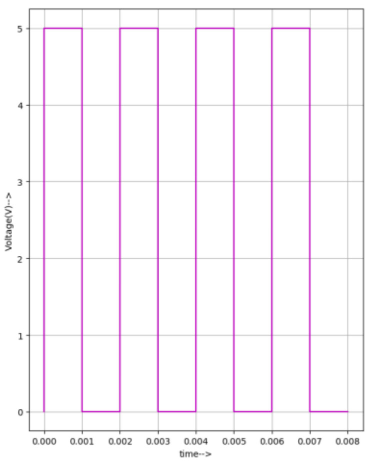

Fig. 8. Input B1

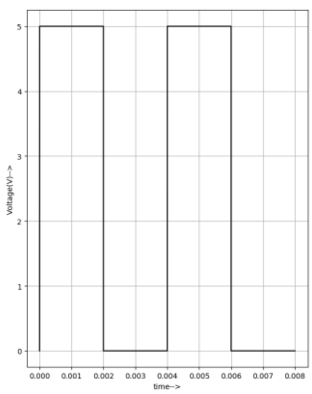

Fig. 9. Input B2

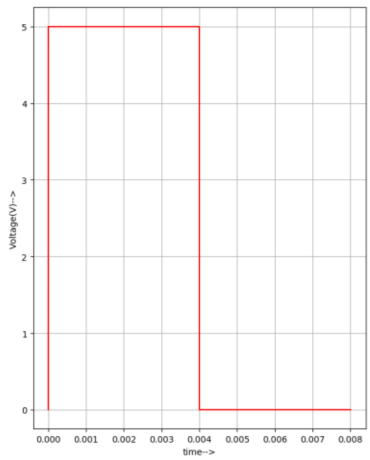

Fig. 10. Input B3

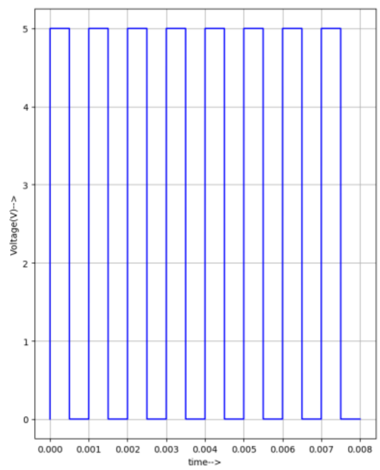

Fig. 11. Input Cin

Figures 3 to 11 represent the four input bits of A, four input bits of B and Cin.

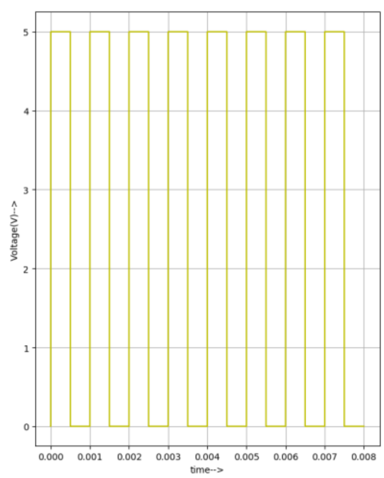

Fig. 12. Output S0

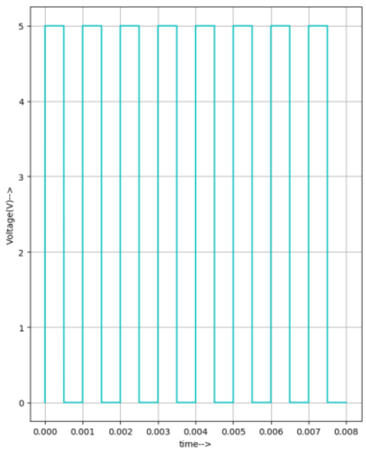

Fig. 13. Output S1

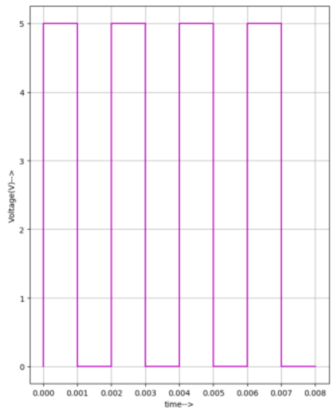

Fig. 14. Output S2

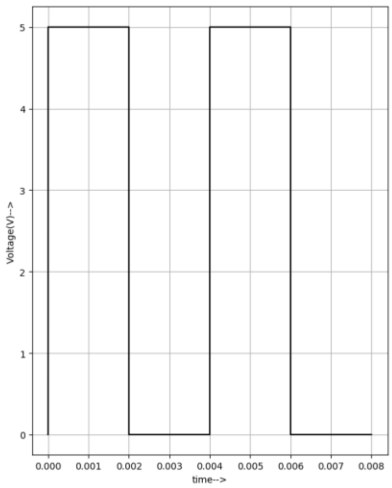

Fig. 15. Output S3

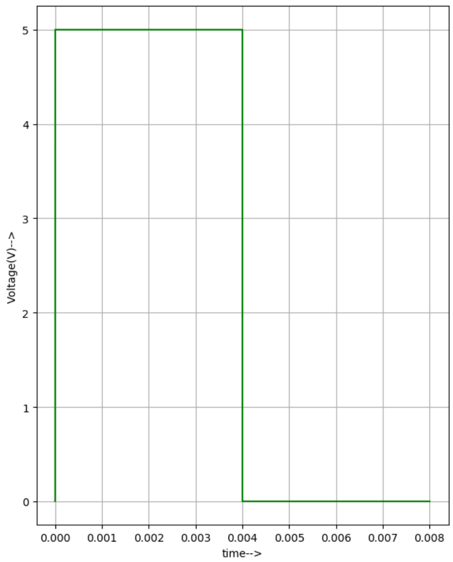

Fig. 16. Output Cout

Figures 12 to 16 represents the sum bits and Cout.

V. Analysis

The working of the CLA was verified using transient analysis. The waveforms displayed the switching of generate, propagate, and carry signals over time. The results confirmed proper timing, accurate carry computation, and faster performance compared to ripple-based adders. The parallel calculation of carries was clearly observed in the simulation.

VI. Conclusion

The 4-bit Carry Lookahead Adder was successfully implemented in eSim using full adders and lookahead logic. Transient analysis confirmed that the circuit produced correct sum outputs (S0–S3) and carry output (C4) for all input combinations. The results showed proper timing behavior, accurate operation, and reduced delay, demonstrating the speed advantage of the CLA in digital systems.

References

Kumar, R., & Dahiya, S. (2013). Performance analysis of different bit carry look ahead adder using VHDL environment. *International Journal of Engineering Science and Innovative Technology (IJESIT)*, 2(4), 1–5. ISSN: 2319-5967.